1.7.4 ▶ ▶ ▶ Bernoulli's Equation

Bernoulli's equation is essentially just conservation of energy applied to a moving fluid. It gives an easy way to calculate properties of an ideal fluid at certain positions. To derive it, first consider the following diagram of an ideal fluid of density ρ in a tube.

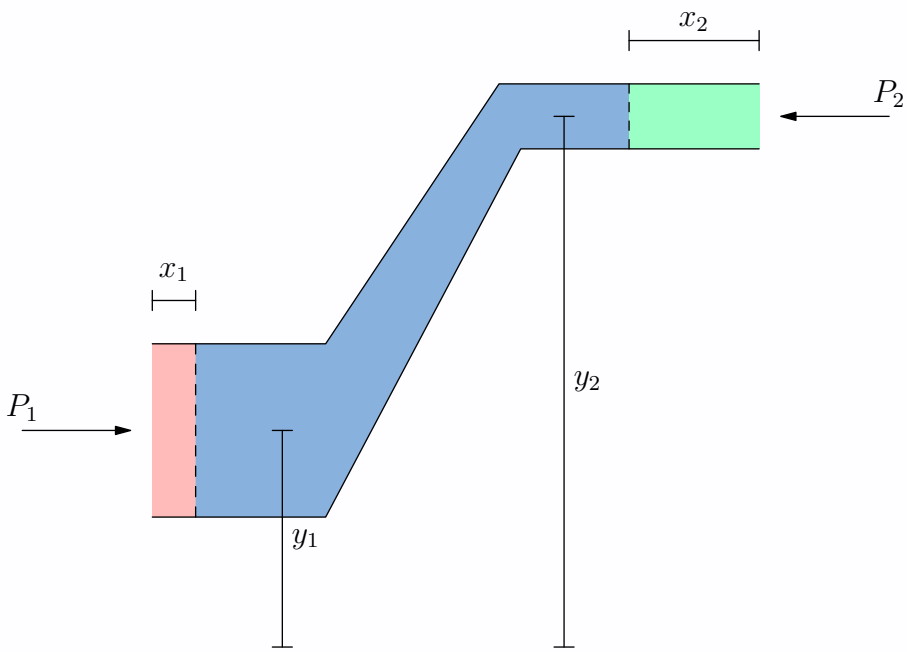

Fig. 1.18

Consider that the tube has cross-sectional area A_1 on the left end and cross-sectional area A_2 on the right. Further consider that the fluid has speed v_1 on the left end and v_2 on the right end. Imagine that fluid fills only the red and blue sections to begin. After some time, the fluid is pushed by the pressure differential created by P_1 and P_2, and the fluid advances to the blue and green sections. If we write the total energy of the fluid in its initial state, we have

$$E_i = \frac{1}{2}mv_1^2 + mgy_1 + E_b$$

where E_b is the total energy of the fluid in the blue section, and m is the mass of the fluid in the red section. After the fluid flows to the blue and green sections, the total energy of the fluid is

$$E_f = \frac{1}{2}mv_2^2 + mgy_2 + E_b.$$

Note that the mass of the fluid in the green section is equal to the mass of the fluid in the red section since the fluid is incompressible and mass is conserved. By the work-energy theorem, any changes in energy occur due to work done by net external forces. In this case, the work is done by the pressures P_1 and P_2. The total work done by these pressures is

$$W_{net} = P_1A_1x_1 - P_2A_2x_2.$$

Since we know that the mass of the fluid in the red green sections are equal, and the fluid

has a constant density, $A_1x_1 = A_2x_2 = \frac{m}{\rho}$, which means that

$$w_{net} = (P_1 - P_2)\frac{m}{\rho}.$$

From work energy theorem, we have

$$W_{net} = E_f - E_i$$

$$(P_1 - P_2)\frac{m}{\rho} = \frac{1}{2}mv_2^2 + mgy_2 - \frac{1}{2}mv_1^2 - mgy_1$$

$$P_1 + \frac{1}{2}\rho v_1^2 + \rho g y_1 = P_2 + \frac{1}{2}\rho v_2^2 + \rho g y_2 \tag{1.116}$$

We can also write the above equation as

$$\boxed{P + \frac{1}{2}\rho v_2 + \rho g y = const} \tag{1.117}$$

which is Bernoulli's equation. Notice that if the height of the fluid doesn't change, pressure has an inverse relation to the kinetic energy. This inverse relation means that faster-moving fluid within a constant-energy stream has lower pressure than slower-moving fluid. This effect, known as the Bernoulli effect, is responsible for the lift force generated in sailboats and airplanes ★